Lahore University of Management Sciences
Syed Babar Ali School of Science and Engineering
AI-500 – Fall 2025 – Mid-term Exam – Part I

Time Allowed: Initial 30 mins **Total Points**: 14+1

+1 mark is for following the instructions

Campus ID: _____ **Signature:** _____

Do Not Open This Exam Unless Told To Do So

Problem 1. Each MCQ is for 1 marks, each MCQ have only one correct answer.

(1) Which of the following data structures is used to implement Breadth-First Search (BFS)?
   (a) Stack
   (b) Queue
   (c) Priority Queue
   (d) Linked List

(2) In BFS, a node is visited:
   (a) Before all of its ancestors
   (b) After all of its ancestors
   (c) Level by level
   (d) In random order

(3) Which traversal strategy does BFS follow?
   (a) Depth-wise traversal
   (b) Level-order traversal
   (c) Random traversal
   (d) Recursive traversal

(4) In a graph, node S has two successors A and B with the following costs: g(S, A)=2, g(S, B)=3, h(A)=4, h(B)=2. Which node will A* expand first?
   (a) A
   (b) B
   (c) Both together
   (d) Depends on tie-breaking

(5) A* search guarantees an optimal path if:
   (a) The heuristic is admissible
   (b) The heuristic is consistent
   (c) Both A and B
   (d) Neither

(6) If A* uses an overestimating heuristic, what happens?
   (a) A. The algorithm will still find the optimal path
   (b) It may miss the optimal path
   (c) It will take longer but still be optimal
   (d) It behaves like BFS

(7) Given a search space where each edge cost = 1, and heuristic h(n) = 0 for all n. What does A* reduce to?

(a) A. Greedy Best-First Search
(b) Uniform Cost Search
(c) Depth-First Search
(d) Hill Climbing

(8) Given beam width = 2 and the following log probabilities after two decoding steps:
Step 1: a: -0.5, b: -0.7
Step 2 expansions: a→c: -1.0, a→d: -0.8, b→c: -1.2, b→d: -0.4
Which two sequences survive after step 2?

(a) a→d, b→d
(b) a→d, a→c
(c) a→c, b→c
(d) b→d, b→c

(9) In beam search if beam width = 1, beam search reduces to:

(a) Greedy Search
(b) Random Search
(c) Exhaustive Search
(d) Monte Carlo Sampling

(10) In an Isolation Forest, how are anomalies detected?

(a) A. By measuring reconstruction error
(b) By counting the number of splits needed to isolate a point
(c) using the distance from cluster centroids
(d) By checking density of neighbors

(11) The Isolation Forest algorithm is based on:

(a) Random partitioning of data
(b) Gradient boosting
(c) Density estimation
(d) Principal Component Analysis

(12) Which of the following statements about Isolation Forest is true?

(a) A. It requires labeled data
(b) It is an unsupervised learning algorithm
(c) It is a type of neural network
(d) It is used only for regression

(13) What is the time complexity of the Minimax algorithm for a game tree of branching factor b and depth d?

(a) $O(b + d)$
(b) $O(bd)$
(c) $O(b^d)$
(d) $O(d^b)$

(14) Which of the following is a limitation of Minimax?

(a) orks only for two-player games
(b) Exponential time complexity
(c) Cannot handle stochasticity
(d) All of the above